R1.4 Entropy and spontaneity (HL)

Gibbs energy and equilibrium (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to perform calculations using the equation $\Delta G = \Delta G^{\ominus} + RT \ln Q$ and its application to a system at equilibrium $\Delta G^{\ominus} = -RT \ln K$.

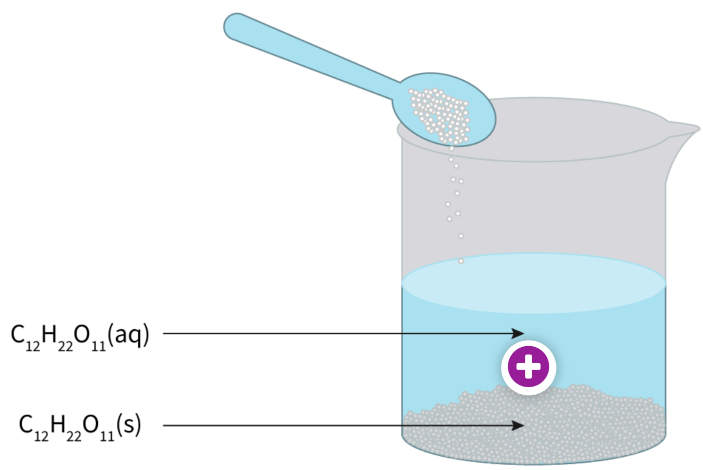
So far, you have learned about various reactions where reactants are transformed into products in unidirectional reactions. For example, when a log of wood burns, the compounds that made up the wood will be irreversibly changed into carbon dioxide and water.

However, there are processes and reactions that can exist with a balance between reactants and products. Think of a time when you have added sugar to your tea. Maybe you noticed that if you add a lot of it, no matter how long you stir, you will not be able to dissolve more of your sugar and you end up with a little pile at the bottom of your cup. What happens is that the system reaches an equilibrium, where the amount of sugar that is dissolving, is the same as the amount of sugar returning to the solid state. This system is said to be in equilibrium and it can be represented by the following equation:

$$C_{12}H_{22}O_{11}(s) \rightleftharpoons C_{12}H_{22}O_{11}(aq)$$

Notice that there is a double-headed half-arrow representing the equilibrium of the process of dissolution of sucrose, $C_{12}H_{22}O_{11}$, which is the compound present in white sugar. What do you think the double-headed arrow implies for this process?

Use **Interactive 1** to investigate how sugar dissolves in tea.



Interactive 1. Solution of Sugar in Tea at Equilibrium.

More information for interactive 1

An interactive image illustrates the process of dissolving sugar in water. A spoonful of solid sugar, $C_{12}H_{22}O_{11}(s)$, is added to a beaker containing water. As the sugar dissolves, it forms an aqueous solution, $(C_{12}H_{22}O_{11}(aq)$, represented by dispersed molecules in the liquid. The image highlights the distinction between undissolved solid sugar at the bottom and dissolved sugar in the solution. The hotspot, when clicked, pops up another image which can be described as follows:
A magnified view of the dissolving process shows individual sugar molecules separating from the solid and dispersing into the solution. Arrows indicate movement, representing the dynamic equilibrium between dissolved molecules and undissolved sugar.

Given what you know now about spontaneity of reactions, you might be wondering:

What happens to ΔG for systems that reach an equilibrium?

Reversible reactions and equilibrium

So far, we have considered that chemical reactions transform reactants into products in a one-way direction. However, there are some reactions that are reversible. That means that the products of the forward reaction can be used as reactants to yield the original compounds (backward reaction.) When that happens we say the system is in equilibrium.

For example, bromine, Br_2, is a liquid at room temperature, but it is very volatile. This means that it can easily turn into a gas. Bromine has a brown colour as you can see in **Video 1**.

Video 1. Bromine, Br_2, being poured and the gas evolving.

When bromine is initially placed in a closed container, some of the particles move from the liquid phase to the gas phase. This is expected, as you know that entropy of the system should increase, which means that the ΔG is negative. You can see the brown liquid at the bottom and a brown gas starting to fill the container.

But after a while, there are no more visible changes to the unaided eye. Once the gas phase has been filled with as many bromine particles as possible in the space available, the number of particles moving in both directions (liquid-to-gas phase and gas-to-liquid phase) is the same. This can be represented as:

$$Br_2(l) \rightleftharpoons Br_2(g)$$

The forward reaction is $Br_2(l) \rightarrow Br_2(g)$, and the backward reaction is $Br_2(g) \rightarrow Br_2(l)$.

Notice again the double-headed arrow (⇌) showing that the system is at equilibrium and be careful not to confuse it with the double headed arrow for resonance structures (⇔).

Figure 1 shows a visual representation of what is happening inside a closed jar with bromine.

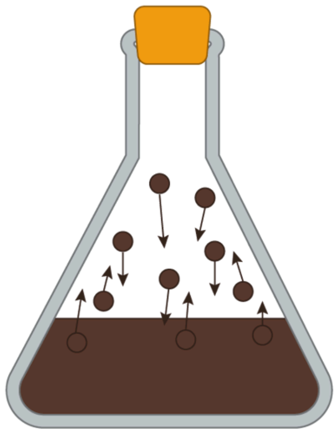


Figure 1. Bromine, Br_2, at equilibrium: the rate of evaporation is equal to the rate of condensation.

More information for figure 1

The image depicts a laboratory flask with a stopper at the top, containing bromine depicted as a brown liquid at the bottom. Within the flask, there are arrows pointing upwards from the liquid, representing the evaporation of bromine molecules into the gas phase, and arrows pointing downward, indicating condensation back into liquid form. This illustrates the dynamic equilibrium where the rate of evaporation equals the rate of condensation.

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Making connections

You can read more about systems in equilibrium in subtopic R2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/).

For reversible reactions, we can determine the extent of the reaction by calculating the equilibrium constant, K, and the reaction quotient Q.

The equilibrium constant is calculated using equilibrium concentrations of reactants and products.

The reaction quotient is calculated using the concentrations of reactants and products when the system is not at equilibrium. For example, for the general reaction:

$$aA + bB \rightleftharpoons cC + dD$$

the equilibrium constant is written as:

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

where a, b, c and d represent the stoichiometric coefficients in the balanced equation. The reaction quotient (Q) is written the same way, but the concentrations of products and reactants used should be the ones when the system is not at equilibrium.

When writing these expressions, there are a few important notes:

- the products are on the top of the expression and the reactants on the bottom
- square brackets [ ] are used to show the concentrations (in mol dm^{-3}) of the reactants and products
- the powers to which the concentrations are raised are the stoichiometric coefficients in the balanced equation for the reaction.

Study skills

Note that the equilibrium constant is temperature dependent. This means that it has different values at different temperatures.

Also note that the magnitude of K is an indication of the extent to which a chemical reaction favours the formation of products or reactants at equilibrium.

- When $K >> 1$ this means that at equilibrium, there are relatively more products than reactants, indicating that the reaction strongly favours the formation of products.
- When $K << 1$, it means that at equilibrium, there are relatively more reactants than products, indicating that the reaction strongly favours the formation of reactants.

Worked example 1

Write an expression for the equilibrium constant, K, for the reaction of sulfur dioxide (SO_2) with oxygen (O_2) to form sulfur trioxide (SO_3):

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

$$K = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

Note that when $Q = K$, the reaction is at equilibrium (**Figure 2**).

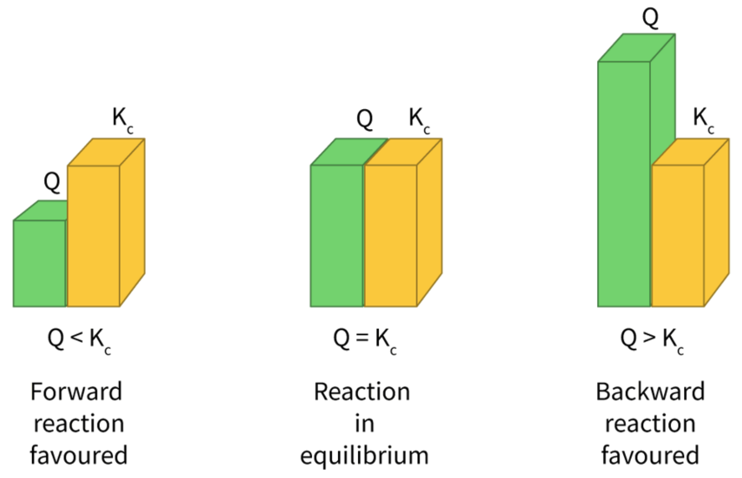

Figure 2. The importance of the relative sizes of Q and K in determining the direction of the reaction to reach equilibrium.

More information for figure 2

The image is a bar chart illustrating the comparative sizes of Q and K, which are variables in a chemical reaction to determine the direction needed to reach equilibrium. The chart consists of three sets of bars. In the first set, the green bar representing Q is shorter than the yellow bar representing K. In the second set, the green bar representing Q and the yellow bar representing K are of equal height, indicating equilibrium. In the third set, the green bar representing Q is taller than the yellow bar representing K. The Y-axis is labeled as "Size."

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Gibbs energy for reactions at equilibrium

The relationship between the Gibbs energy change (ΔG) and the spontaneity of a chemical reaction was covered in section R1.4.2-3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/gibbs-energy-and-spontaneous-changes-hl-id-44766/). Recall that, for a spontaneous reaction, the Gibbs energy change must be negative.

We can use the following equation in **Table 1** to calculate the Gibbs energy change for a reaction not at equilibrium (note the use of the reaction quotient, Q).

Table 1. Equation for Gibbs energy change.

Equation	Symbols	Units
$\Delta G = \Delta G^{\ominus} + RT \ln Q$	ΔG = the Gibbs energy change	kilojoules per mc mol^{-1})
	$\Delta G^{\ominus}$ is the standard Gibbs energy change for the reaction	kilojoules per n (kJ mol^{-1})
	R is the universal gas constant (8.31 J K^{-1} mol^{-1})	joules per kelvir mole (J K^{-1} mc

Equation	Symbols	Units
	T = the absolute temperature	kelvin (K)
	$\ln Q$ = the natural logarithm of the reaction quotient	no units

Note that R has the units J K^{-1} mol^{-1}. To convert to kJ mol^{-1}, divide by 1000.

Worked example 2

Consider the reaction to produce ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The standard Gibbs energy, $\Delta G^{\ominus}$, is –32.7 kJ mol^{-1} and the reaction quotient, Q, is 6.4×10^{-7}.

Determine the change in Gibbs energy, ΔG, for this reaction when $T = 100\,°C$.

Solution steps	Calculations
Step 1: Write out the information provided in the question.	$\Delta G^{\ominus} = -32.7\ \text{kJ mol}^{-1}$ $Q = 6.4 \times 10^{-7}$
Step 2: Convert any values that need changing.	$T = 100 + 273$ $= 373\ \text{K}$
Step 3: Write out the equation.	$\Delta G = \Delta G^{\ominus} + RT \ln Q$
Step 4 Substitute the values given.	$= -32.7 + \left(\frac{8.31}{1000}\right)(373)\ln(6.4\times10^{-7}) =$ Note that the gas constant, R, has been divid K^{-1} mol^{-1}.
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$\Delta G = -77\,\text{kJ mol}^{-1}\ (2\ \text{s.f.})$ Given that ΔG is negative, this means favoured at $T = 100\,°\text{C}$ (which will mo equilibrium).

The standard Gibbs energy change, $\Delta G^{\ominus}$, for a reversible reaction can also be determined for systems at equilibrium taking into consideration the equilibrium constant, K, using the equation in **Table 2**.

Table 2. Equation for standard Gibbs energy change.

Equation	Symbols	Units
$\Delta G^{\ominus} = -RT \ln K$	$\Delta G^{\ominus}$ is the standard Gibbs energy change for the reaction	joules per m (J mol^{-1})
	R is the universal gas constant (8.31 J K^{-1} mol^{-1})	joules per kelvi mole (J K^{-1} m
	T = the absolute temperature	kelvin (K)
	$\ln K$ = the natural logarithm of the equilibrium constant	

Note that R has the units J K^{-1} mol^{-1}, so $\Delta G^{\ominus}$ in this equation has units of J mol^{-1}. To convert to kJ mol^{-1}, divide by 1000.

From the equation above, you can notice that ΔG is directly proportional to the natural logarithm of K. This means that when a reaction favours the formation of products at equilibrium (large K), the forward reaction is more spontaneous (ΔG negative). When the reaction has a low value of K, it means that the reaction favours the formation of reactants at equilibrium, and the ΔG change for the reaction is less negative (the forward reaction is less spontaneous) or even positive (the backward reaction is spontaneous).

Worked example 3

Consider again the reaction used to produce ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The standard Gibbs energy, $\Delta G^{\ominus}$, is –32.7 kJ mol^{-1} and T = 25 °C.

Determine the equilibrium constant, K.

Solution steps	Calculations
Step 1: Write out the information provided in the question.	$\Delta G^{\ominus}$, is -32.7 kJ mol^{-1} $T = 25\ °C$
Step 2: Convert any values that need changing.	$T = 25 + 273$ $= 298$ K
Step 3: Write out the equation.	$\Delta G^{\ominus} = -RT \ln K$

Solution steps	Calculations
Step 4 Substitute the values given.	$\ln K = -\frac{\Delta G^{\ominus}}{RT}$ $= -\frac{-32.7}{\left(\frac{8.31}{1000}\right)(298)}$ $= 13.2$ $K = e^{13.2}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$= 5.4 \times 10^5$ (2 s.f.) Note that as $K >> 1$, the forwa favoured and, at equilibrium, th concentrations of the products

Study skills

The equations that relate Gibbs energy, ΔG, with the extent of the reaction (Q or K) for a reversible reaction are given in the DP chemistry data booklet in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/).

$$\Delta G = \Delta G^{\ominus} + RT \ln Q$$

$$\Delta G^{\ominus} = -RT \ln K$$

As a reversible reaction proceeds, the composition of the reaction mixture is constantly changing, as is the Gibbs energy. The position of equilibrium corresponds to a maximum value of entropy (S) and a minimum value of Gibbs energy change (ΔG). Once this point has been reached, the reaction will not proceed any further – the rates of the forward reaction and the reverse reaction are now equal. Which means the reaction quotient, Q, is equal to the equilibrium constant, K, ($Q = K$).

By substituting the first equation into the second we get:

$$\Delta G = -RT \ln K + RT \ln Q$$

Therefore:

$$\Delta G = RT \left(\ln \frac{Q}{K}\right)$$

Recall, that at equilibrium, $Q = K$, therefore, $\frac{Q}{K} = 1$. The natural log (ln) of $1 = 0$, so:

$$\Delta G = RT(\ln 1) = 0$$

From this, we can see that at equilibrium, when $Q = K$, the ΔG for a reaction is zero.

Note that the standard change in Gibbs energy ($\Delta G^{\ominus}$) only has a value of zero when the equilibrium constant, K, is equal to 1:

$$\Delta G^{\ominus} = -RT\ln(1) = 0$$

Figure 3 shows that when the equilibrium constant, K, has a value of 1, the value of the $\Delta G^{\ominus}$ is zero. For such a reaction, this means that neither the products nor reactants are favoured at equilibrium.

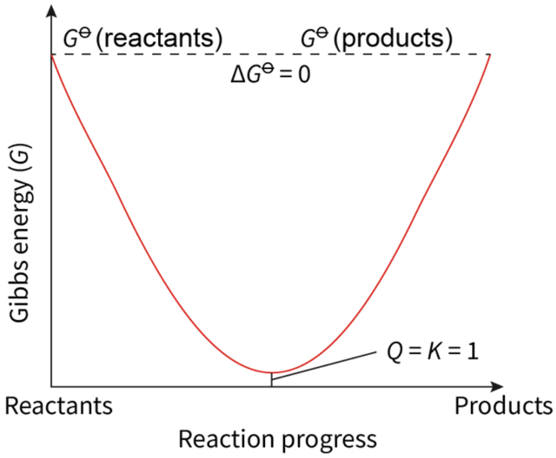

Figure 3. The standard change in Gibbs energy ($\Delta G^{\ominus}$) only has a value of zero when the equilibrium constant, K, is equal to 1

More information for figure 3

The graph is titled with measures of Gibbs energy (G) against the reaction progress, depicted with a curve. The X-axis represents 'Reaction progress', starting from 'Reactants' on the left to 'Products' on the right. The Y-axis is labeled as 'Gibbs energy (G)'. The curve displays an initial high Gibbs energy at the reactants, decreasing to a minimum point before ascending again to the product side, illustrating a concave-upward U-shaped curve.

Key points of the graph include labels: (G^\circ) (reactants) and (G^\circ) (products), with a dashed line running parallel to the X-axis at these points. The text '(\Delta G^\circ = 0)' is positioned between the dashed lines for reactants and products, suggesting that the standard change in Gibbs energy is zero at equilibrium. An annotation on the curve highlights 'Q = K = 1,' indicating the equilibrium condition where the reaction is perfectly balanced between products and reactants.

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Gibbs energy and spontaneity of a reversible reaction

Given the equations studied above, you can use the values of standard Gibbs energy change, $\Delta G^{\ominus}$, and the equilibrium constant, K, to determine if the forward reaction of a reversible reaction is spontaneous or not.

Compare the values in **Table 3** that shows K and $\Delta G^{\ominus}$ for various chemical reactions.

Table 3. Values of $\Delta G^{\ominus}$ and K for selected reactions at 298 K.

Reaction	$\Delta G^{\ominus}$ (kJ mol^{-1})	*K*
$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$	+142	1.4 × 10
$H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$	+79.9	1.0 × 10
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$	−32.9	5.8 ×
$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$	−211	1.4 × 1

You might notice that the more negative values of standard Gibbs energy change, $\Delta G^{\ominus}$, mean larger values of the equilibrium constant, K. This means that the more spontaneous a reaction is, the further the position of equilibrium lies to the right (i.e. the forward reaction (products) is favoured).

Use **Interactive 3** to summarise the relationship between the equilibrium constant, K, and the standard Gibbs energy change, $\Delta G^{\ominus}$.

$\Delta G^{\ominus}$	ln *K*	*K*	Position of equilibrium
Negative			
	Zero		
		< 1	

Negative | Positive | Zero | < 1 | = 1 | > 1

Lies to the left – favouring the reactants | Lies to the right – favouring the products | Equal proportions of reactants and products

Check

Interactive 3. Summarise the relationship between K, and $\Delta G^{\ominus}$.

When the position of the equilibrium lies to the right on a reversible reaction, this means that the forward reaction is spontaneous. For example, for the reactions listed in the interactive:

- The reaction between zinc and aqueous copper ions has a large negative value for the $\Delta G^{\ominus}$ and a very large value for the equilibrium constant, K. From this, we can deduce that the reaction goes almost to completion, with the equilibrium mixture containing mostly products.

- The decomposition of sulfur trioxide, SO_3, has a large positive value for the $\Delta G^{\ominus}$ and a very small value for the equilibrium constant, K. From this, we can deduce that, at 298K, the forward reaction barely proceeds, and the equilibrium mixture contains mostly reactants.

Worked example 4

Consider the reaction to produce ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

You calculated the value of Gibbs energy, $\Delta G^{\ominus} = -72.5\ \text{kJ mol}^{-1}$ and the equilibrium constant, $K = 5.4 \times 10^5$.

Determine if the reaction favours the products or reactants.

Given that $\Delta G < 0$ and $K > 1$, the forward reaction is spontaneous and so the products are favoured at 25 °C (298 K).

Figure 4 shows how the Gibbs energy changes for spontaneous and non-spontaneous reactions. For both, the Gibbs energy decreases until it reaches a minimum value, when the reaction is at equilibrium. This point also corresponds to the point of maximum entropy for the reaction, because mixtures have higher entropy than pure substances.

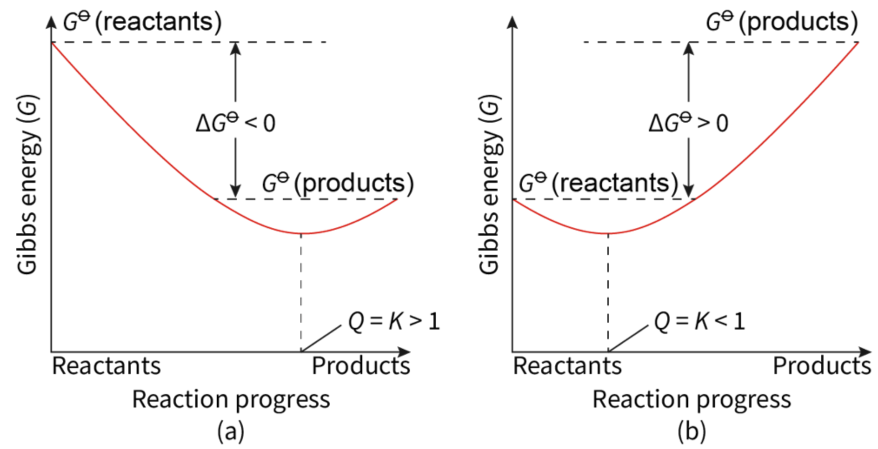

Figure 4. Plot of Gibbs energy versus reaction progress for spontaneous (a) and non-spontaneous (b) reactions.

More information for figure 4

The image contains two graphs depicting the relationship between Gibbs energy and reaction progress for spontaneous and non-spontaneous reactions.

On the left (a): The graph shows a curve that starts high at the Reactants side and decreases towards the Products side, indicating a spontaneous reaction. The X-axis represents Reaction progress with labels for Reactants on the left and Products on the right. The Y-axis represents Gibbs energy (G), with the notation G° for reactants and products. The curve dips to a minimum point at equilibrium, marked with G° (products) lower than G° (reactants). ΔG° is less than 0, and Q equals K which is greater than 1.

On the right (b): This graph illustrates a non-spontaneous reaction where the curve rises as it moves from Reactants to Products. The X-axis also represents Reaction progress, and the Y-axis is the Gibbs energy (G). The energy at G° (products) is higher than G° (reactants), with ΔG° greater than 0, and Q equals K, which is less than 1. The minimum energy point is positioned closer to Reactants, signifying higher reactants at equilibrium.

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For a spontaneous reaction, the position of equilibrium lies towards the products side. This is shown by the position of the minimum value of the Gibbs energy, which is closer to the products side than the reactants side, i.e. there is a higher concentration of products than reactants at equilibrium, with a K value greater than 1.

For a non-spontaneous reaction, this point on the graph lies more towards the reactants side than the products side. This means that at equilibrium, there is a greater concentration of reactants than products in the reaction mixture, with a K value of less than 1.

Making connections

Subtopic R2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/) What is the likely composition of an equilibrium mixture when $\Delta G^{\ominus}$ is positive?

For a reversible reaction, you can analyse the relationship between the Gibbs energy and extent of the reaction through the equation:

$$\Delta G^{\ominus} = -RT \ln K$$

When ΔG is positive, this means that the forward reaction is not spontaneous under the given conditions and that energy must be supplied to drive the reaction forward. In this case, K<<1. In a hypothetical reversible reaction:

$$aA + bB \rightleftharpoons cC + dD$$

the equilibrium constant is written as:

$$K = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

This means that for a reaction where K<<1, the concentration of products is much lower than the concentration of reactants at equilibrium.

Try the activity in the next section to investigate the effect of temperature on reaching equilibrium.